



Green University of Bangladesh
Department of Computer Science and Engineering (CSE)
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Sorting Algorithm Visualizer

Project Proposal

Course Title: Algorithms Lab
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Chapter 1

Introduction

1.1 Overview

Sorting Algorithm Visualizer is a software project developed using the C programming language. The main purpose of this project is to show how different sorting algorithms work step by step. Many students find sorting algorithms difficult because they cannot see what happens during the sorting process. This project solves that problem by showing the movement of data while sorting.

The program will allow users to choose a sorting algorithm from a menu. After selecting an algorithm, the program will display each step of the sorting process. The user can compare different algorithms and understand their working process easily. The project includes the following sorting algorithms:

- Bubble Sort
- Selection Sort
- Insertion Sort
- Merge Sort
- Quick Sort

1.2 Motivation

Sorting is one of the most important topics in computer science. Students usually learn sorting by reading code, but it is hard to understand how the algorithms actually work.

The motivation of this project is to make learning easier by showing the sorting process visually. When users can watch every comparison and swap, they understand the algorithm better. This project also helps students compare the speed and behavior of different sorting methods.

1.3 Problem Definition

1.3.1 Problem Statement

Many students have difficulty understanding sorting algorithms because they only see the program code or final output. They cannot easily understand how the data changes during sorting.

This project creates a visual learning tool that displays each step of the sorting process. It helps users learn, compare, and understand different sorting algorithms in a simple way. The project needs to solve several technical problems:

- Display the sorting process step by step.
- Update the screen after every comparison and swap.
- Handle different sorting algorithms in one program.
- Keep the animation smooth and easy to understand.
- Make the program simple and user-friendly.

1.3.2 Complex Engineering Problem

Table 1.1: Summary of the attributes touched by the mentioned projects

Name of the P Attributes	Explain how to address
P1: Depth of knowledge required	Knowledge of C programming, arrays, loops, functions, and sorting algorithms is required.
P2: Range of conflicting requirements	The program should be simple to use while also showing clear visualization
P3: Depth of analysis required	Compare different sorting algorithms based on their working process and efficiency.
P4: Familiarity of issues	Understand common problems like swapping elements, comparisons, and execution time.
P5: Extent of applicable codes	Use standard C programming and basic console functions.
P6: Extent of stakeholder involvement and conflicting requirements	Students and teachers need an easy learning tool with clear visualization.
P7: Interdependence	The visualization depends on correct implementation of each sorting algorithm and screen updates.

1.4 Design Goals/Objectives

The main objectives of this project are:

- To develop a sorting algorithm visualizer using C.
- To visualize sorting algorithms step by step.
- To help students understand sorting algorithms easily.
- To compare different sorting algorithms.
- To create a simple and user-friendly interface.
- To improve programming and algorithm learning.

1.5 Application

This project can be used in many areas, such as:

- Teaching sorting algorithms in programming classes.
- Learning data structures and algorithms.
- Laboratory demonstrations.
- Self-learning for beginner programmers.
- Comparing different sorting algorithms for educational purposes.